

PG 1

- ① GIVEN FUEL ROD $r = 0.55 \text{ cm}$ $k_f = 0.025 \text{ W/cm}\cdot\text{K}$
 $\text{LHR} = 300 \text{ W/cm}$ $E = 200 \text{ GPa}$ ← WATCH UNITS!
 $\nu = 0.35$ $\alpha = 10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$

a. MAX STRESS:

NEED $(T_o - T_s)$ FOR $\sigma_r = \frac{\alpha E (T_o - T_s)}{4(1-\nu)}$ ✓

$200 \text{ GPa} = 200,000 \text{ MPa}$

$T_o - T_s = \frac{\text{LHR}}{4\pi k} = \frac{300 \text{ W/cm}}{4\pi(0.025 \text{ W/cm}\cdot\text{K})} = \frac{300}{4\pi(0.025)} = 954.9 \text{ K}$

$\sigma_r = \frac{10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1} (954.9 \text{ K}) (200,000 \text{ MPa})}{4(1-0.35)} = 734.54 \text{ MPa}$

$\sigma_{\text{max}} = \sigma_r(r) = -\sigma_r \left(\frac{1-3\eta^2}{1+\eta^2} \right)$ ASSUME SURFACE $\eta = 1$

$\Rightarrow -\sigma_r (1-3(1)^2) = -\sigma_r (-2) = 2\sigma_r$

$\sigma_{\text{max}} = 2(734.54 \text{ MPa}) = \boxed{1469.1 \text{ MPa}}$

b. CALC. CRACKS INTO FUEL:

GIVEN: $\sigma_{\text{fract}} = 175 \text{ MPa}$

$\sigma_r(r) = \sigma_r (3\eta^2 - 1)$ SETTING $\sigma_r = 175 \text{ MPa}$

$\Rightarrow 175 \text{ MPa} = 734.54 \text{ MPa} (3\eta^2 - 1)$

$\Rightarrow 3\eta^2 - 1 = \frac{175 \text{ MPa}}{734.54 \text{ MPa}} + 1$

$3\eta^2 = 1.238$

$\eta = \sqrt{\frac{1.238}{3}}$

$\eta = 0.642$

radius of tip $= R_r(\eta) = 0.55 \text{ cm} (0.642) = 0.353 \text{ cm}$

$a = R_r = R(1-\eta) = 0.55(1-0.642) = \boxed{0.197 \text{ cm}}$

PG 2

② GIVEN (ROD PRESS @ 15MPa
(R_{avg}) AVG RADIUS: 0.55cm
CHD THICK: 0.05cm

a. CALC ALL THREE USING THIN WALL CYL.

$$\sigma_t = \frac{PR}{\delta} = \frac{15MPa(0.55)}{0.05} = \boxed{165MPa}$$

$$\sigma_z = \frac{PR}{2\delta} = \frac{15MPa(0.55)}{2(0.05)} = \boxed{82.5MPa}$$

$$\sigma_r = -\frac{P}{2} - \frac{15MPa}{2} = \boxed{-7.5MPa}$$

b. CALC USING THICK WALL CYL.

NEED $R_i + R_o + \delta$

$$R_i = R_{avg} - \frac{\delta}{2} = 0.55 - 0.025 = 0.525cm$$

$$R_o = R_{avg} + \frac{\delta}{2} = 0.55 + 0.025 = 0.575cm$$

$$D = \left(\frac{R_o}{R_i}\right)^2 - 1 = \left(\frac{0.575}{0.525}\right)^2 - 1$$

$$D = 0.200$$

@ R_i
FIX

$$\sigma_r(r) = \frac{-P\left(\frac{R_o}{r}\right)^2 - 1}{\left(\frac{R_o}{r_i}\right)^2 - 1} = -15MPa \left[\frac{\left(\frac{0.575}{0.525}\right)^2 - 1}{\left(\frac{0.575}{0.525}\right)^2 - 1} \right] =$$

$$-15MPa \left[\frac{0.0929}{0.1995} \right] =$$

$$\sigma_r(R_i) = -P \left[\frac{\left(\frac{R_o}{R_i}\right)^2 - 1}{\left(\frac{R_o}{R_i}\right)^2 - 1} \right] \Rightarrow \sigma_r(R_i) = -P = \boxed{-15MPa}$$

$$\sigma_\theta(r) = P \left[\frac{\left(\frac{R_o}{r}\right)^2 + 1}{\left(\frac{R_o}{R_i}\right)^2 - 1} \right] \Rightarrow \sigma_\theta(R_i) = P \left[\frac{\left(\frac{R_o}{R_i}\right)^2 + 1}{\left(\frac{R_o}{R_i}\right)^2 - 1} \right] = \left[\frac{\left(\frac{0.575}{0.525}\right)^2 + 1}{\left(\frac{0.575}{0.525}\right)^2 - 1} \right]$$

$$\sigma_\theta(R_i) = \boxed{165.3MPa}$$

$$\sigma_z(r) = P \left[\frac{\left(\frac{R_o}{r}\right)^2 - 1}{\left(\frac{R_o}{R_i}\right)^2 - 1} \right] = 15MPa \left[\frac{\left(\frac{0.575}{0.525}\right)^2 - 1}{\left(\frac{0.575}{0.525}\right)^2 - 1} \right] = \boxed{75.2MPa}$$

PG 3

③ CALC Δ GAP $\checkmark$ DUE TO T_{EXP} .

GIVEN: $R_f = 0.48 \text{ cm}$ $T_{\infty} = 550 \text{ K}$
 $t_{gap} = 0.008 \text{ cm}$ $t_{clad} = 0.08 \text{ cm}$
 $k_{fuel} = 0.03 \text{ W/cm-K}$ $k_{gap} = 0.003 \text{ W/cm-K}$
 $k_{clad} = 0.15 \text{ W/cm-K}$ $L + r = 250 \text{ W/cm}$
 $\alpha_c = 10 \times 10^{-6} \text{ 1/K}$ $T_{ref} = 300 \text{ K}$
 $\alpha_f = 14 \times 10^{-6} \text{ 1/K}$ (FUEL = CLAD)

$$\Delta \sqrt{gap} = \bar{R}_c \alpha_c (\bar{T}_c - T_{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref})$$

NEED: $\bar{T}_c, \bar{T}_c, T_o, T_s, \bar{R}_c$

ASSUME $\bar{T}_{oc} = T_{\infty} = 550 \text{ K}$

$$T_{ic} - T_{oc} = \frac{L + r}{2\pi R_f (h_{clad})} = \frac{250 \text{ W/cm}}{2\pi (0.48) (h_{clad})}$$

$$T_{ic} = \frac{250}{2\pi (0.48) (1.875)} + T_{oc}$$

$$\Delta T_{gap} \quad T_{ic} = 44.21 \text{ K} + 550 \text{ K} = 594.21 \text{ K}$$

$$T_s - T_{ic} = \frac{L + r}{2\pi R_f (h_{gap})}$$

$$T_s = \frac{250}{2\pi (0.48) (0.375)} + T_{ic}$$

$$T_s = 221.05 \text{ K} + 594.21 \text{ K} = 815.26 \text{ K}$$

$$\Delta_{fuel} \quad T_o - T_s = \frac{L + r}{4\pi k_f} \Rightarrow T_o = \frac{250}{4\pi (0.03)} + T_s$$

$$T_o = 663.15 \text{ K} + 815.26 \text{ K} = 1478.41 \text{ K}$$

$$h_{clad} = \frac{k_{clad}}{t_{clad}} = \frac{0.15 \text{ W/cm-K}}{0.08 \text{ cm}} = 1.875 \text{ cm}^2 \cdot \text{K}$$

$$h_{gap} = \frac{k_{gap}}{t_{gap}} = \frac{0.003 \text{ W/cm-K}}{0.008 \text{ cm}} = 0.375 \text{ cm}^2 \cdot \text{K}$$

$$R_c = R_f + t_{gap} = 0.48 \text{ cm} + 0.008 \text{ cm} = 0.488$$

NEED $\bar{T}_f, \bar{T}_c$

$$\bar{T}_f = T_s + \frac{\Delta_{fuel}}{2} = 815.26 + \left(\frac{663.15}{2}\right) = 1146.84 \text{ K}$$

$$\bar{T}_c = \frac{T_{ic} + T_{oc}}{2} = \frac{594.21 \text{ K} + 550 \text{ K}}{2} = 572.11 \text{ K}$$

$$\Delta \sqrt{gap} = 0.488 (10 \times 10^{-6} \text{ 1/K}) (572.11 \text{ K} - 300 \text{ K}) - 0.48 (14 \times 10^{-6} \text{ 1/K}) (1146.84 \text{ K} - 300 \text{ K})$$

$$\Delta \sqrt{gap} = -0.0044 \text{ cm} \Rightarrow \text{NEW GAP IS } 0.0036$$

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- ④ FIND σ + r STRESS @ INNER RADIUS OF CLADDING

GIVEN: $\Delta T = 50K$ $\alpha_c = 15 \times 10^{-6}$
 UNITS! $E = 100 GPa$ $\nu = 0.34$
 $t_c = 0.06 cm$ $R_1 = 0.55 cm$

RADIAL: IF $r = R_1$, THEN $\frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left[\frac{r}{R_1} - 1 \right] \left[1 - \frac{R_1^2}{r^2} \left(\frac{r}{R_1} - 1 \right) \right]$
 $\Rightarrow \sigma_r(R_1) = 0$

$\sigma_\theta(R_1) = \frac{1}{2} \Delta T \left[\frac{\alpha E}{1-\nu} \right] \left[1 - 2 \frac{R_1^2}{r^2} \left(\frac{r}{R_1} - 1 \right) \right]$
 $\frac{1}{2} \Delta T \left[\frac{\alpha E}{1-\nu} \right] [1]$

$\Rightarrow \sigma_\theta(R_1) = \frac{1}{2} [50K] \left[\frac{15 \times 10^{-6} \cdot 100,000 MPa}{1 - 0.34} \right]$

$\sigma_\theta(R_1) = 56.82 MPa$

- ⑤ FIND DENSIFICATION

GIVEN: FISSION RATE: 6×10^{13} fissions/cm³-s

$\Delta \rho_0 = 0.015$

$\beta_0 = 5 MW/D / kg U$

$C_0 = 1$

$E_D = \Delta \rho_0 \left(e^{\frac{\beta_0 \Delta \rho_0}{C_0 \beta_0}} - 1 \right)$

✓ convert!

$\rho(UO_2) = 10.97 g/cc$

$t = 100 days \rightarrow$

$t = 864000 s$

$\frac{100 \text{ days}}{1 \text{ day}} \cdot \frac{24 \text{ hr}}{1 \text{ hr}} \cdot \frac{60 \text{ min}}{1 \text{ min}} \cdot \frac{60 \text{ s}}{1 \text{ s}} = 864000 s$

NEED: β

$\beta = \frac{\dot{F}(t)}{N_0} = \frac{1.922 \times 10^3 W/cm^3 (864000 s)}{6 \times 10^{13} fissions/cm^3 \cdot s (864000 s)}$
 $\rightarrow 0.00967 kgU/cm^3$

$E_D / f_{iss} = 200 MeV = 3.204 \times 10^{-11} J$

$\Rightarrow 6 \times 10^{13} fissions/cm^3 \cdot s \cdot 3.204 \times 10^{-11} J/f_{iss} = 1.922 \times 10^3 W/cm^3$

$\frac{238}{238.02891} \approx 0.99987$ $\rho_{UO_2} = 10.97 g/cc$
 $\rho_U = 0.967 gU/cm^3 = 0.00967 kgU/cm^3$

$\beta = 1.717 \times 10^{-4} J/kgU = 1.99 MW/D / kgU$

Pg 5

⑤ CONT

$$\epsilon_D = \Delta \rho_0 \left(\frac{13 \ln(6.01)}{\cos 30} - 1 \right)$$

$$= 0.015 \left(\frac{1.99 \ln(6.01)}{21.5} - 1 \right)$$

$$\epsilon_D = -0.013 \Rightarrow \approx 1.3\% \text{ DENSIFICATION}$$

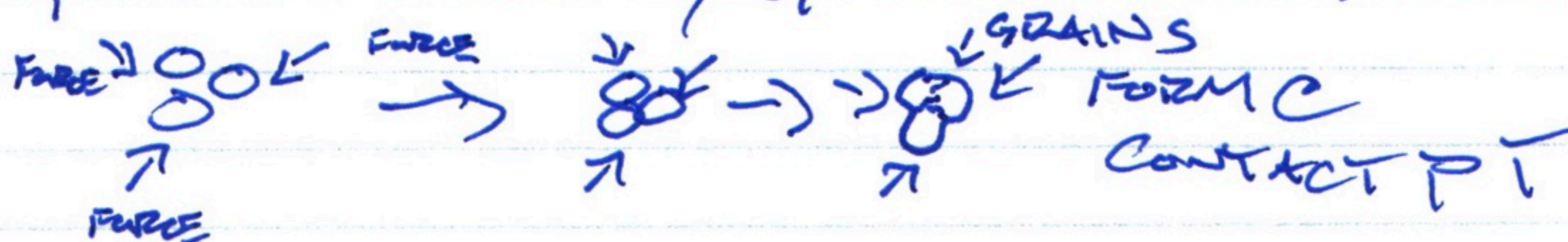
⑥ ELASTICITY IS THE ABILITY OF AN OBJECT TO RETURN TO ORIGINAL STATE AFTER A LOAD IS APPLIED

PLASTICITY IS AN IRREVERSIBLE Δ AFTER YIELDING TO A LOAD

⑦ STRAIN HARDENING IS WHEN A MATERIAL IS TAKEN INTO THE PLASTIC DEFORMATION REGION, BUT BELOW THE TEMP OF RECRYSTALLIZATION.
 γ + HARDNESS $\uparrow\uparrow$, BUT DUCTILITY $\downarrow$

⑧ A THERMAL ANALYSIS, MECHANICAL ANALYSIS, AND MATERL BEHAVIORS OVER TIME.

⑨ DURING SINTERING, POWDER PARTICLES BOND @ CONTACT POINTS CREATING GRAIN BOUNDARIES



⑩ ~~MESH~~ USING MICROSTRUCTURE MODELING KNOWS PER ANALYSIS OF GRAIN SIZE + BOUNDARY INTERACTIONS. IT ALSO ALLOWS FOR A BETTER UNDERSTANDING OF DIFFUSION AND GAS BUBBLE GROWTH.

ALL THESE GIVE BETTER PREDICTIONS + REDUCE UNCERTAINTY.

3/6/26

Pg 6

- (11) MICROSTRUCTURE IS THE INTERNAL MATERIAL STRUCTURE AT THE GRAIN OR PORE LEVEL OF A MATERIAL.

AN EXAMPLE IS SINTERING BY Δ TEMP OR TIME DURING PROCESS, THE MICROSTRUCTURES COULD BECOME LARGER OR SMALLER $\Rightarrow$ AING THE MATERIALS OVERALL PROPERTIES. THIS COULD INCLUDE BETTER OR WORSE GAS DIFFUSION CHARACTERISTICS.

- (12) DIFFUSION WILL OCCUR FASTER @ GRAIN BOUNDARIES DUE TO HAVING A BETTER HIGHER ENERGY STATE FROM STRESS @ BOUNDARY + DEFECTS ALLOWING BETTER GAS MOVEMENT THEN AN ORDERED LATTICE.

- (13) GRAINS CAN GROW DUE TO $\uparrow$ TEMP IN SYSTEM, LONGER TIME @ TEMPS, $\downarrow$ PINNING (IMPURITY) MATERIALS
GRAINS CAN SLOW IN GROWTH DUE TO PINNED BOUNDARIES
OR RUNNING @ LOWER TEMPS

- (14) FUEL WILL DENSIFY BECAUSE THE SINTERING PROCESS CREATES PORES/VOIDS IN PELLET. AS $T \uparrow$ THESE PORES SHRINK AND MATERIAL DENSITY $\uparrow$ WITH DIMENSIONS OF THE FUEL REDUCING.


PORE IN
VOL
FUEL @ GRAIN
BOUNDARIES.